

MAT 549 HW01

Aditya Pahuja

Due 02/10

Problem 1. Let M be a nonempty positive-dimensional smooth manifold with or without boundary. Show that $\mathfrak{X}(M)$ is infinite-dimensional.

Solution. By working in local coordinates, we may assume that $M = \mathbb{R}^n$. For $k \in \mathbb{N}$, let $f_k: M \rightarrow [0, \infty)$ such that $f^{-1}(0) = \mathbb{R}^n - \overline{B}_{1/k}(0)$ (e.g. $f(x) = 1/k^2 - |x|^2$ for $x \in B_{1/k}(0)$ and $f(x) = 0$ otherwise); that is, the supports of the f_k are a strictly decreasing sequence of sets. Define the vector fields $X_k = f_k \frac{\partial}{\partial x_1}$.

Given the vector fields $X_{i_1}, \dots, X_{i_m}$ with $i_1 < \dots < i_m$ that are linearly dependent, consider a linear combination $a_1 X_{i_1} + \dots + a_m X_{i_m}$ for nonzero real $a_1, \dots, a_m$. By construction, X_{i_1} is nonzero at some points outside $\overline{B}_{1/i_2}(0)$ while all the other vector fields are zero outside this closed ball, so the linear combination is nonzero on $B_{1/i_1}(0) - \overline{B}_{1/i_2}(0)$. Thus any finite linear combination of the X_k is nonzero, meaning $\{X_1, X_2, \dots\}$ is an infinite linearly independent subset of $\mathfrak{X}(M)$, so the space of vector fields on M is infinite-dimensional. $\square$

Problem 2. Let M be a smooth manifold with boundary. Show that there exists a global smooth vector field on M whose restriction to ∂M is everywhere inward-pointing, and one whose restriction to ∂M is everywhere outward-pointing.

Solution. Let $(U_\alpha, \varphi_\alpha)_{\alpha \in I}$ be a collection of charts on M whose domains cover M , and let $(\psi_\alpha)_{\alpha \in I}$ be a partition of unity subordinate to the covering $(U_\alpha)_{\alpha \in I}$. Then, for each $\alpha \in I$, define a vector field X_α on U_α as follows: if U_α is contained in the interior of M , then set $X_\alpha \equiv 0$; otherwise, define $X_\alpha = \frac{\partial}{\partial x_n}$ in local coordinates, so that $X_\alpha|_{\partial M \cap U_\alpha}$ is an inward-pointing vector at each point of $\partial M \cap U_\alpha$. Then,

$$X = \sum_{\alpha \in I} \varphi_\alpha X_\alpha$$

is a global vector field for which $X|_{\partial M}(p)$ is the sum of some inward-pointing vectors, hence inward-pointing, for every $p \in \partial M$. Negating X yields a global vector field that is outward-pointing at every point of ∂M . $\square$

Problem 3. For each of the following vector fields on the plane, compute its coordinate representation in polar coordinates on the right half-plane.

(a) $X = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$

(b) $Y = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}$

(c) $Z = (x^2 + y^2) \frac{\partial}{\partial x}$.

Solution. By the change of coordinates formula (equation (3.11)),

$$\frac{\partial}{\partial r} = \cos \theta \cdot \frac{\partial}{\partial x} + \sin \theta \cdot \frac{\partial}{\partial y}, \quad \frac{\partial}{\partial \theta} = -r \sin \theta \cdot \frac{\partial}{\partial x} + r \cos \theta \cdot \frac{\partial}{\partial y}.$$

This implies that

$$X = r \cos \theta \cdot \frac{\partial}{\partial x} + r \sin \theta \cdot \frac{\partial}{\partial y} = \boxed{r \frac{\partial}{\partial r}}, \quad Y = -r \sin \theta \cdot \frac{\partial}{\partial x} + r \cos \theta \cdot \frac{\partial}{\partial y} = \boxed{\frac{\partial}{\partial \theta}}$$

and we can solve the system of equations for $\frac{\partial}{\partial x}$ directly to get

$$Z = \boxed{r^2 \cos \theta \cdot \frac{\partial}{\partial r} - r \sin \theta \cdot \frac{\partial}{\partial \theta}}. \quad \square$$

Problem 4. Show that there is a smooth vector field on S^2 that vanishes at exactly one point.

Solution. Consider the vector field $X \in \mathfrak{X}(S^2)$ given by

$$X(x, y, z) = \frac{1}{2}(y^2 - x^2 + (1 - z)^2, -2xy, 2x(1 - z)).$$

The component functions are smooth on $\mathbb{R}^3$, hence of course smooth on S^2 . The vectors are tangent to S^2 because

$$\begin{aligned} X(x, y, z) \cdot (x, y, z) &= \frac{1}{2}(xy^2 - x^3 + x(1 - z)^2 - 2xy^2 + 2xz(1 - z)) \\ &= \frac{1}{2}(-x^3 - xy^2 + x(1 - z^2)) \\ &= \frac{x}{2}(1 - x^2 - y^2 - z^2) = 0. \end{aligned}$$

Finally, to see that the vector field is only zero at one point, observe that $X(x, y, z) = 0$ implies that $x = 0$ or $y = 0$ by looking at the second component; the first case means the first component $y^2 + (1 - z)^2 = 0$, hence $y = 0$ and $1 - z = 0$; on the other hand, the second case (assuming $x \neq 0$) implies that $1 - z = 0$ from the third component and then $x = 0$ because the first component equals $x^2 = 0$. In either case, $X(x, y, z) = 0$ implies $(x, y, z) = (0, 0, 1)$, so X is nonzero away from $(x, y, z) = (0, 0, 1)$. $\square$

Problem 5. For each of the following pairs of vector fields, compute the Lie bracket $[X, Y]$.

(a) $X = y \frac{\partial}{\partial z} - 2xy^2 \frac{\partial}{\partial y}; Y = \frac{\partial}{\partial y}$

(b) $X = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}; Y = y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}$

(c) $X = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}; Y = x \frac{\partial}{\partial y} + y \frac{\partial}{\partial x}$

Solution. (a)

$$[X, Y] = X(1) \frac{\partial}{\partial y} - Y(-2xy^2) \frac{\partial}{\partial y} - Y(y) \frac{\partial}{\partial z} = \boxed{4xy \frac{\partial}{\partial y} - \frac{\partial}{\partial z}}$$

(b)

$$[X, Y] = X(-z) \frac{\partial}{\partial y} + X(y) \frac{\partial}{\partial z} - Y(-y) \frac{\partial}{\partial x} - Y(x) \frac{\partial}{\partial y} = \boxed{x \frac{\partial}{\partial z} - z \frac{\partial}{\partial x}}$$

(c)

$$[X, Y] = X(x) \frac{\partial}{\partial y} + X(y) \frac{\partial}{\partial x} - Y(-y) \frac{\partial}{\partial x} - Y(x) \frac{\partial}{\partial y} = \boxed{2x \frac{\partial}{\partial x} - 2y \frac{\partial}{\partial y}} \quad \square$$